

# Introduction to Probabilistic Models

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# Motivation



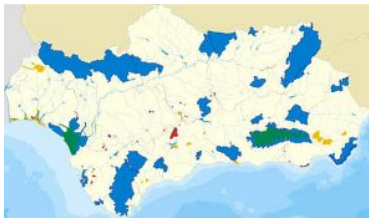
*I always wondered how it would be if  
a Superior species landed on Earth and  
showed us how they played chess.  
Now I know it.*

Peter Heine Nielsen  
Chess Grand Master and Magnus Carlsen's coach

The question is,

- Can we (**humans**) learn (**interpret**) anything from it?

# Examples



Monitoring protected areas



Matching victims' DNA to relatives



Predicting events to avoid



Gen AI

# Probabilistic models

## All the previous examples:

- Operate in environments where large amounts of data are available
- However, **data don't cover all the possible scenarios**  $\Rightarrow$  **UNCERTAINTY**
- Use a **probabilistic model**, typically learnt from data
- Use inference algorithms to carry out **prediction** and **structure analysis**

## Probabilistic models offer at least:

- Principled quantification of uncertainty
- Natural way of dealing with missing data
- **Interpretability**

## What we also need from them:

- Ability to operate in **high dimensional** spaces
- Support **efficient** inference and learning

# Uncertainty

We often (try to) distinguish between two types of uncertainty:

- **Epistemic:** Due to lack of knowledge
- **Aleatoric:** Due to (pure) randomness, i.e. the variability in the outcome of an experiment due to random effects

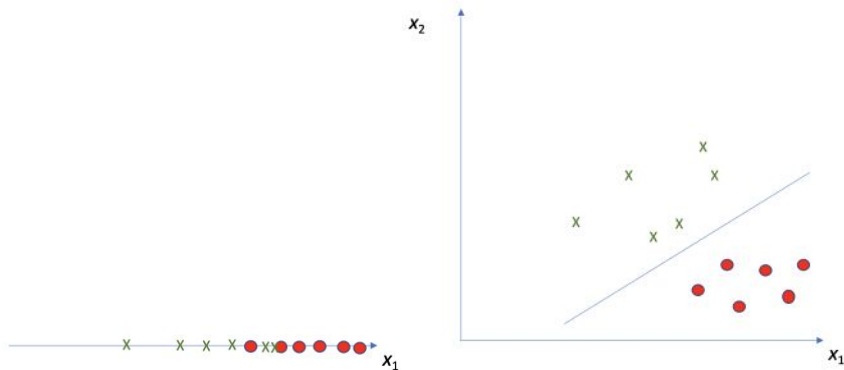
## Example

- Assume we want to predict  $Y$  from  $X$
- We estimate a joint distribution  $p(x, y)$  [EPISTEMIC][REDUCIBLE]
- We predict  $Y$  using  $p(y|x) = p(x, y)/p(x)$
- If we observe  $X = x$ , what does our model predict for  $Y$ ?  
[ALEATORIC][IRREDUCIBLE]

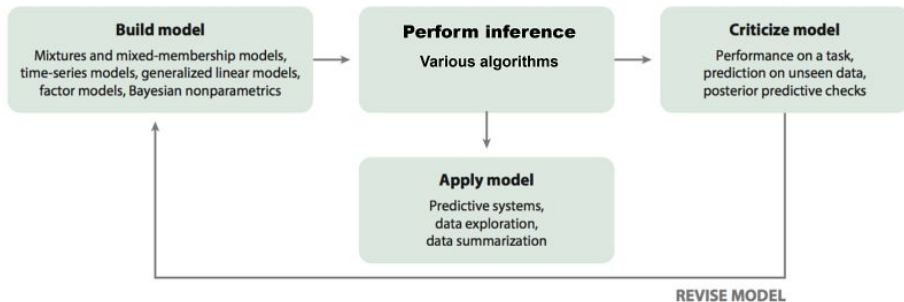
# Uncertainty

Epistemic uncertainty can be reduced by

- gathering more data, but also by **increasing the number of features**

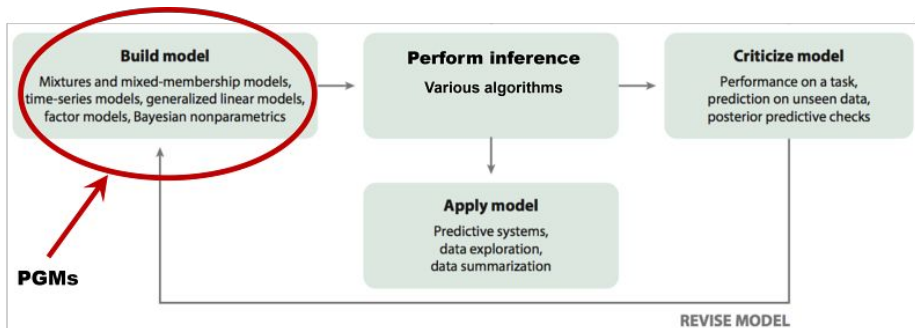


# The Probabilistic Modelling Cycle



Trade-offs (performance, richness, costs); **be clear about assumptions!**

# The Probabilistic Modelling Cycle



## Probabilistic graphical models offer:

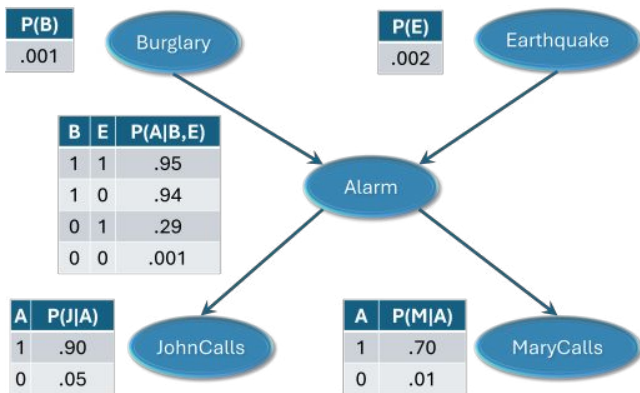
- **Structured** specification of high dimensional distributions in terms of low dimensional factors
- **Efficient** inference and learning, taking advantage of the structure
- **Graphical** representation interpretable by humans



# Bayesian network: definition

A **Bayesian network** over random variables  $\mathbf{X} = \{X_1, \dots, X_n\}$  consists of

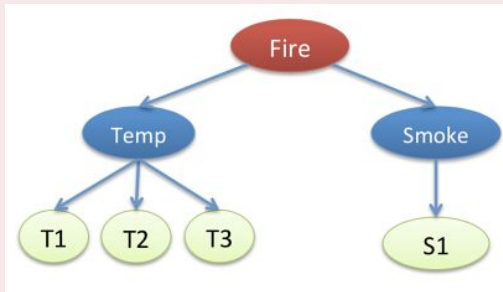
- A **DAG**  $\mathcal{G} = (\mathcal{V}, \mathcal{E})$  with  $\mathcal{V} = \mathbf{X}$
- A set of **local conditional distributions**  $\mathcal{P} = \{ p(X_i \mid pa(X_i)) \mid X_i \in \mathbf{X} \}$  where  $pa(X_i)$  denotes the parents of  $X_i$  according to  $\mathcal{E}$



# Bayesian network: compact representation of the joint

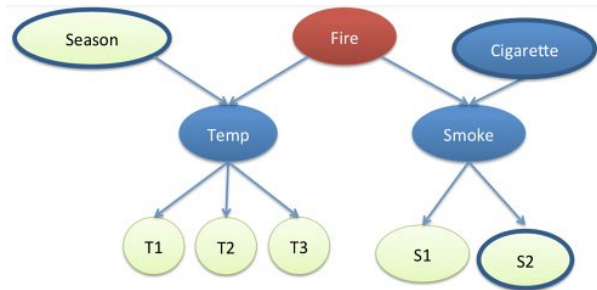
The DAG encodes **independences** among the variables (through **d-separation**); every Bayesian network therefore represents a **factorized** joint distribution:

$$p(X_1, \dots, X_n) = \prod_{i=1}^n p(X_i \mid pa(X_i))$$



$$p(f, t, s, t_1, t_2, t_3, s_1) = p(t_1|t)p(t_2|t)p(t_3|t)p(s_1|s)p(t|f)p(s|f)p(f)$$

# Bayesian networks: modular structure



$$p(\textcolor{red}{se}, f, \textcolor{blue}{c}, t, s, t_1, t_2, t_3, s_1) = p(t_1|t)p(t_2|t)p(t_3|t)p(s_1|s)\textcolor{red}{p}(s_2|s) \\ p(t|\textcolor{red}{se}, f)p(s|f, \textcolor{blue}{c})p(\textcolor{red}{se})p(\textcolor{blue}{f})p(c)$$

# Inference in Bayesian networks

Assume a Bayesian network over variables  $\mathbf{X} = \{X_1, \dots, X_n\}$

From the **joint distribution**  $p(X_1, \dots, X_n)$  we can infer a.o.

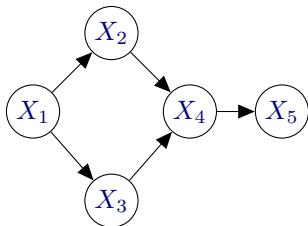
- the **prior distribution**  $p(X_i)$  of any  $X_i$ ,
- the **posterior distribution**  $p(X_i \mid \mathbf{x}_E)$  of any  $X_i$  given **evidence** for  $\mathbf{x}_E$ .

Note: interpretation of these terms will differ when we consider **learning**!

## Inference methods

- **Exact**
  - Brute force: compute  $P(\mathbf{X}, \mathbf{x}_E)$  and marginalize out  $\mathbf{X} \setminus \mathbf{X}_I$
  - Take advantage of the network structure
- **Approximate**
  - Sampling
  - Deterministic

## Exact inference: Variable elimination



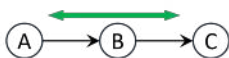
- We are interested in  $X_5$
- All variables are discrete
- $E = \emptyset$

$$\begin{aligned} p(x_5) &= \sum_{\mathbf{x}_1, \dots, \mathbf{x}_4} p(\mathbf{x}_1, x_2, x_3, x_4, x_5) \\ &= \sum_{\mathbf{x}_1, \dots, \mathbf{x}_4} p(\mathbf{x}_1) p(x_2 | \mathbf{x}_1) p(x_3 | \mathbf{x}_1) p(x_4 | x_2, x_3) p(x_5 | x_4) \\ &= \sum_{x_2, \dots, x_4} \sum_{\mathbf{x}_1} p(\mathbf{x}_1) p(x_2 | \mathbf{x}_1) p(x_3 | \mathbf{x}_1) p(x_4 | x_2, x_3) p(x_5 | x_4) \\ &= \sum_{x_2, \dots, x_4} p(x_4 | x_2, x_3) p(x_5 | x_4) \boxed{\sum_{\mathbf{x}_1} p(\mathbf{x}_1) p(x_2 | \mathbf{x}_1) p(x_3 | \mathbf{x}_1)} \\ &= \sum_{x_2, \dots, x_4} p(x_4 | x_2, x_3) p(x_5 | x_4) h(x_2, x_3) \end{aligned}$$

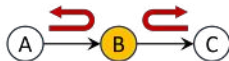
We have reached a similar problem as initially, but with **one variable less**.

# Interpreting Bayesian network structures: $d$ -separation

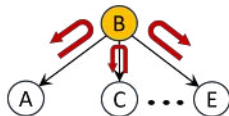
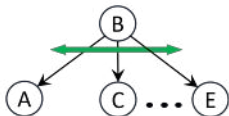
Active



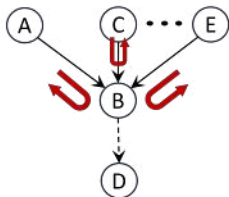
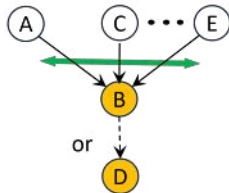
Blocked



- Serial connection

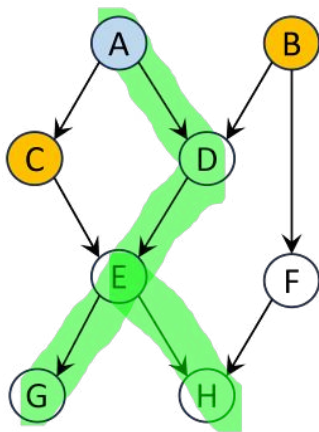


- Diverging connection



- Converging connection

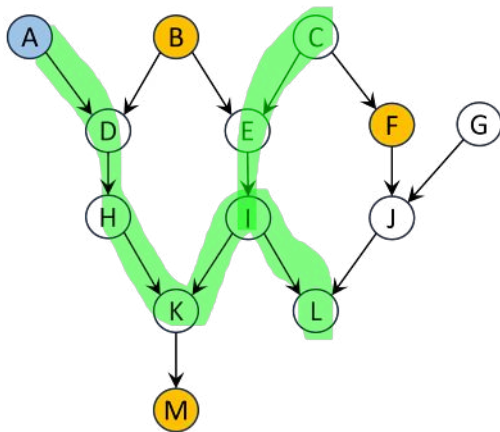
## $d$ -separation example - I



Which variables are  $d$ -separated from  $A$  given the evidence (in orange)?

All outside active (green) chains  $\Rightarrow A \perp F \mid B, C$

## $d$ -separation example - II

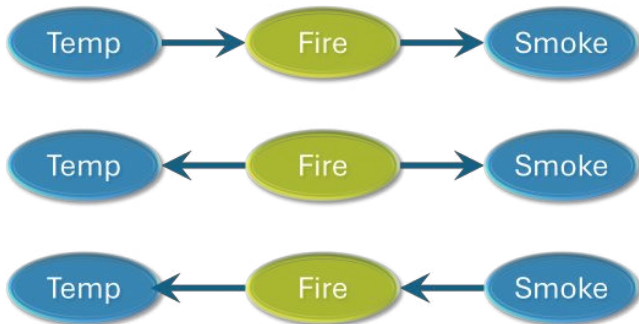


Which variables are  $d$ -separated from  $A$  given the evidence (in orange)?

All outside active (green) chains  $\Rightarrow A \perp G, J \mid B, F, M$



# What about causality?



- All are (Markov) **equivalent**, from an **observational** perspective
- What if I **cause a fire** rather than observe a fire?

# Monty Hall problem

You are given the choice between 3 doors. One has a real prize behind it, the other two joke prizes.

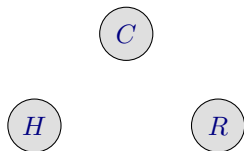


You choose a door; the host then opens a door and offers you the choice to switch to a closed door.

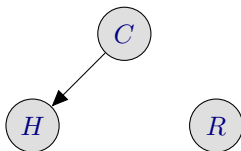
Would you switch?

# Monty Hall problem: which DAG fits your assumptions?

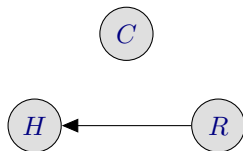
$C$ : your choice of door;  $H$ : door opened by host Monty;  $R$ : door with real prize



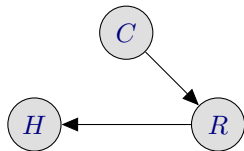
(I)



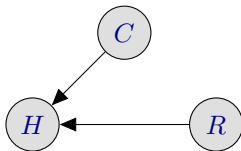
(II)



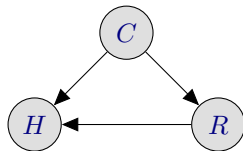
(III)



(IV)



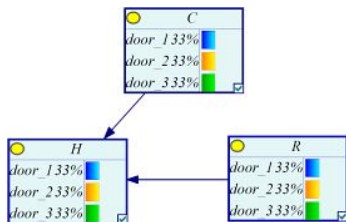
(V)



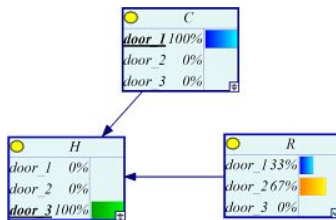
(VI)

- Get the DIY-MontyHall Jupyter notebook (e.g. through Google Colab) from <https://github.com/probabilisticai/probai-2026>

# Monty Hall: performing inference



$$p(H) = \sum_{c,r} p(H | c,r)p(c)p(r)$$



$$p(R | C = \text{door}_1, H = \text{door}_3) = \frac{p(H = \text{door}_3 | C = \text{door}_1, R)p(R)}{p(H = \text{door}_3)}$$

- What assumptions underlie your probability assessments?

# Probabilistic graphical models

**Recall** that probabilistic graphical models offer:

- Structured specification of high dimensional distributions in terms of low dimensional factors
- Graphical representation interpretable by humans
- Efficient inference and learning, taking advantage of the structure

# The probabilistic modelling cycle - II

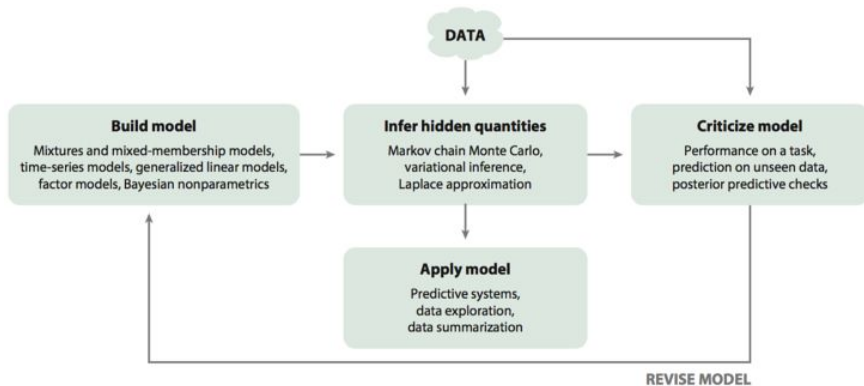


Image from: David M. Blei (2014) "Build, compute, critique, repeat: Data analysis with latent variable models." *Annual Review of Statistics and its Applications* 1, 303–323.

# Probabilistic Machine Learning

What is machine learning?

*"We say that a computer program  $P$   
learns from experience  $E$   
with respect to some class of tasks  $T$   
and a performance measure  $R$ ,  
if its performance on the tasks in  $T$ , measured in terms of  $R$ , improves  
with experience  $E$ ".  
(Tom Mitchell, 1997)*

# Easy example: linear regression

- We have data about two variables  $X$  and  $Y$  (“experience”)
- We want to predict the value of  $Y$  from the value of  $X$
- To solve this task, we decide to use a **linear regression model**

$$\hat{y} = a + bx$$

- As *performance measure*, we use

$$\text{rmse}(\mathbf{y}, \hat{\mathbf{y}}) = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

with  $\mathbf{y} = \{y_1, \dots, y_n\}$  denoting the data and  $\hat{\mathbf{y}} = \{\hat{y}_1, \dots, \hat{y}_n\}$  denoting the model estimates

Is this really ML???



## Easy example: linear regression

- The linear regression model we have used **IS NOT** a probabilistic model
- We'll see later how it can be approached from a probabilistic point of view

# Learning probabilistic models from data

Model (simple):

- a theoretical **probability density/mass function**  $f$ 
  - associated with **random variable**  $X$
  - having **parameter**  $\theta$

Learning problem:

- We assume  $f$  is known except for parameter  $\theta$
- This is denoted as  $f(x; \theta)$  or  $f(x | \theta)$
- Goal: estimate  $\theta$

Tools:

- for a sample  $X_1, \dots, X_n$  drawn from  $f(x | \theta)$ , the **likelihood function** is:

$$l(\theta | x_1, \dots, x_n) \stackrel{\text{def}}{=} f(x_1, \dots, x_n | \theta) = \prod_{i=1}^n f(x_i | \theta)$$

i.e. the joint density/mass regarded as a **function of parameter**  $\theta$

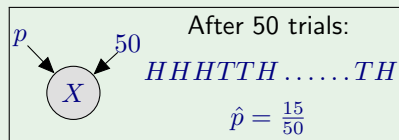
# Learning parameters from data: frequentist approach

- POV: parameter  $\theta$  has a **fixed but unknown** value

**Consider tossing a (fair?) coin**

Goal: estimate  $p(\text{heads})$

Frequentist POV:  
probability = relative frequency  
“in the long run”

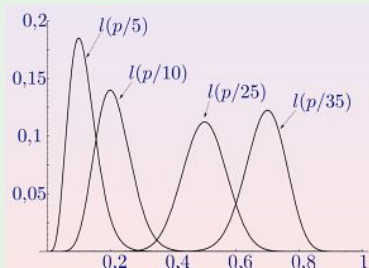


What is underlying theoretical model  $f(x | p)$ ?

Assume a sample of size 1,  
 $X \sim \mathcal{B}(50, p)$

The likelihood function is

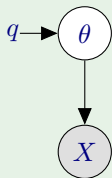
$$l(p | x) = \binom{50}{x} p^x (1 - p)^{50-x}$$



# Learning parameters from data: Bayesian approach

- POV: parameters are modelled as random variables → we can include information prior to observing data
- Additional tools: using Bayes' rule, the prior information is combined with the likelihood, yielding a posterior distribution
- The posterior then becomes the new prior
- As such, inferences about the parameter allow for its updating

## Bayesian networks for Bayesian learning



- Random variables (and parameters) inside circles
- Grey if observable; white if hidden
- Fixed quantities without circle

# Learning from data: Bayesian approach

## Distributions in a Bayesian model - I

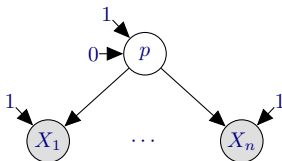
For learning:

- The prior distribution of  $\theta$ ,  $\pi(\theta)$
- The joint distribution of  $(X, \theta)$ ,  $\psi(x, \theta) = f(x|\theta)\pi(\theta)$
- The posterior distribution of  $\theta$  given  $x$ ,

$$\pi(\theta|x) = \frac{f(x|\theta)\pi(\theta)}{\int_{\theta} f(x|\theta)\pi(\theta) d\theta}$$

# Learning from data: Example of Bayesian approach

- Assume a sample  $X_1, X_2, \dots, X_n \sim \mathcal{B}(1, p)$  and  $p \sim \mathcal{U}(0, 1)$



- Then the **likelihood** and the **prior** are,

$$f(x_1, \dots, x_n | p) = p^{\sum x_i} (1 - p)^{n - \sum x_i}, \quad \text{with } x_i = 0, 1; \quad p \in (0, 1),$$

$$\pi(p) = \frac{1}{1 - 0} = 1, \quad \text{if } p \in (0, 1)$$

# Learning from data: Example of Bayesian approach

Assume a sample  $X_1, X_2, \dots, X_n \sim \mathcal{B}(1, p)$  and  $p \sim \mathcal{U}(0, 1)$

- Recall that the **likelihood** and the **prior** are:

$$\begin{aligned} f(x_1, \dots, x_n | p) &= p^{\sum x_i} (1 - p)^{n - \sum x_i}, \quad \text{with } x_i = 0, 1; \quad p \in (0, 1), \\ \pi(p) &= 1, \quad \text{if } p \in (0, 1) \end{aligned}$$

- The **posterior** distribution is

$$\pi(p | x_1, \dots, x_n) = \frac{f(x_1, \dots, x_n | p) \pi(p)}{\int_0^1 f(x_1, \dots, x_n | p) \pi(p) dp} = \frac{p^{\sum x_i} (1 - p)^{n - \sum x_i}}{\int_0^1 p^{\sum x_i} (1 - p)^{n - \sum x_i} dp}$$

# Learning from data: Example of Bayesian approach

## Pattern matching: the Beta distribution $Be(\alpha, \beta)$

$$f(p) = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} p^{\alpha-1} (1-p)^{\beta-1}; \quad \int_0^1 f(p) dp = 1$$

$$\begin{aligned} \int_0^1 p^{\sum x_i} (1-p)^{n-\sum x_i} dp &= \\ &= \int_0^1 \frac{\Gamma(\sum x_i + 1) \Gamma(n - \sum x_i + 1)}{\Gamma(n+2)} \frac{\Gamma(n+2)}{\Gamma(\sum x_i + 1) \Gamma(n - \sum x_i + 1)} p^{\sum x_i} (1-p)^{n-\sum x_i} dp \\ &= \frac{\Gamma(\sum x_i + 1) \Gamma(n - \sum x_i + 1)}{\Gamma(n+2)} \int_0^1 \frac{\Gamma(n+2)}{\Gamma(\sum x_i + 1) \Gamma(n - \sum x_i + 1)} p^{\sum x_i} (1-p)^{n-\sum x_i} dp \\ &= \frac{\Gamma(\sum x_i + 1) \Gamma(n - \sum x_i + 1)}{\Gamma(n+2)} \cdot 1 \end{aligned}$$



# Learning from data: Example of Bayesian approach

Assume a sample  $X_1, X_2, \dots, X_n \sim \mathcal{B}(1, p)$  and  $p \sim \mathcal{U}(0, 1) = Be(1, 1)$

- Then the **likelihood** and the **prior** are,

$$f(x_1, \dots, x_n | p) = p^{\sum x_i} (1 - p)^{n - \sum x_i}, \quad \text{with } x_i = 0, 1; \quad p \in (0, 1),$$
$$\pi(p) = 1, \quad \text{if } p \in (0, 1)$$

- The **posterior** distribution is

$$\begin{aligned} \pi(p | x_1, \dots, x_n) &= \frac{f(x_1, \dots, x_n | p) \pi(p)}{\int_0^1 f(x_1, \dots, x_n | p) \pi(p) dp} = \frac{p^{\sum x_i} (1 - p)^{n - \sum x_i}}{\int_0^1 p^{\sum x_i} (1 - p)^{n - \sum x_i} dp} \\ &= \frac{\Gamma(n + 2)}{\Gamma(\sum x_i + 1) \Gamma(n - \sum x_i + 1)} p^{\sum x_i} (1 - p)^{n - \sum x_i} \end{aligned}$$

which corresponds to  $Be\left(\sum x_i + 1, n - \sum x_i + 1\right)$

Very easy to compute for some models

# Demo

## Bayesian updating

A coin with unknown bias  $\theta$ . Start with a uniform prior. Flip to observe data and watch the posterior update.

Flip once

Flip  $\times 10$

Flip  $\times 50$

Reset

True  $\theta$ :  0.50

Flips

0

Heads

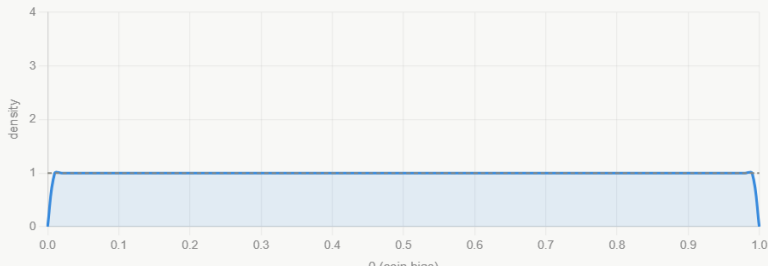
0

Tails

0

Posterior mode

—



# Conjugate priors and likelihoods

Prior and likelihood are called **conjugate**, if prior and posterior are from same family.

| Likelihood                                   | Prior                            | Posterior                                                                             |
|----------------------------------------------|----------------------------------|---------------------------------------------------------------------------------------|
| $\mathcal{B}(1, \theta)$                     | $Be(\alpha, \beta)$              | $Be(\alpha + \sum_{i=1}^n x_i, \beta + n - \sum_{i=1}^n x_i)$                         |
| $\mathcal{NB}(r, \theta)$                    | $Be(\alpha, \beta)$              | $Be(\alpha + rn, \beta - nr + \sum_{i=1}^n x_i)$                                      |
| $\mathcal{G}(\theta)$                        | $Be(\alpha, \beta)$              | $Be(\alpha + n, \beta + \sum_{i=1}^n x_i)$                                            |
| $\mathcal{MN}(n, \theta_1, \dots, \theta_k)$ | $Dir(\alpha_1, \dots, \alpha_k)$ | $Dir(\alpha_1 + x_1, \dots, \alpha_k + x_k)$                                          |
| $P(\theta)$                                  | $\Gamma(\alpha, \beta)$          | $\Gamma(\alpha + \sum_{i=1}^n x_i, \beta + n)$                                        |
| $Exp(\theta)$                                | $\Gamma(\alpha, \beta)$          | $\Gamma(\alpha + n, \beta + \sum_{i=1}^n x_i)$                                        |
| $\mathcal{N}(\mu, \underline{\tau})$         | $\mathcal{N}(\mu_0, \tau_0)$     | $\mathcal{N}(\frac{\tau_0 \mu_0 + n \tau \bar{x}}{\tau_0 + n \tau}, \tau_0 + n \tau)$ |
| $\mathcal{N}(\underline{\mu}, \tau)$         | $\Gamma(\alpha_0, \beta_0)$      | $\Gamma(\alpha_0 + \frac{n}{2}, \beta_0 + \frac{1}{2} \sum_{i=1}^n (x_i - \mu)^2)$    |

# Learning from data: Bayesian approach

## Distributions in a Bayesian model - II

For validation and use (see [Bayesian Workflow](#)):

- The **prior predictive distribution** of  $X$ ,

$$m(x) = \int_{\theta} f(x|\theta)\pi(\theta) d\theta$$

- The **(posterior) predictive distribution** given  $\mathbf{x} = \{x_1, \dots, x_n\}$ :

$$f(x_{n+1}|\mathbf{x}) = \int_{\theta} f(x_{n+1}|\theta, \mathbf{x})\pi(\theta|\mathbf{x})d\theta = \int_{\theta} f(x_{n+1}|\theta)\pi(\theta|\mathbf{x})d\theta$$

## Example Bayesian approach, continued

- The **prior predictive** distribution is

$$m(x) = \int_0^1 p^x (1-p)^{1-x} dp = \frac{\Gamma(x+1)\Gamma(2-x)}{\Gamma(3)} = \frac{x!(1-x)!}{2} = \boxed{\frac{1}{2}} \quad \text{with } x = 0, 1$$

- The **(posterior) predictive** distribution is

$$\begin{aligned} f(x|x_1, \dots, x_n) &= \\ &= \int_0^1 p^x (1-p)^{1-x} \frac{\Gamma(n+2)}{\Gamma(\sum x_i + 1)\Gamma(n - \sum x_i + 1)} p^{\sum x_i} (1-p)^{n - \sum x_i} dp \\ &= \frac{\Gamma(n+2)}{\Gamma(\sum x_i + 1)\Gamma(n - \sum x_i + 1)} \int_0^1 p^{x + \sum x_i} (1-p)^{n+1 - (x + \sum x_i)} dp \\ &= \frac{\Gamma(n+2)}{\Gamma(\sum x_i + 1)\Gamma(n - \sum x_i + 1)} \frac{\Gamma(x+1 + \sum x_i)\Gamma(n+2 - (x + \sum x_i))}{\Gamma(n+3)} \end{aligned}$$

# Learning from data: Bayesian approach

- The method above is known as *fully Bayesian* approach
- Sometimes, we don't need to compute the denominator of the posterior distribution, in which case  $\theta$  can be estimated as

$$\begin{aligned}\hat{\theta} &= \arg \max_{\theta} f(x_1, \dots, x_n, \theta) \\ &= \arg \max_{\theta} f(x_1, \dots, x_n | \theta) \pi(\theta) \\ &= \arg \max_{\theta} \{ \log f(x_1, \dots, x_n | \theta) + \log \pi(\theta) \}\end{aligned}$$

known as the **MAP (Maximum A Posteriori)** estimator

- Note that we could also choose

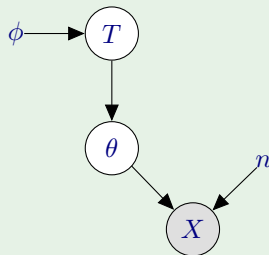
$$\hat{\theta} = \arg \max_{\theta} \log f(x_1, \dots, x_n | \theta)$$

which is actually the **MLE (Maximum Likelihood Estimator)**

# Some simple examples

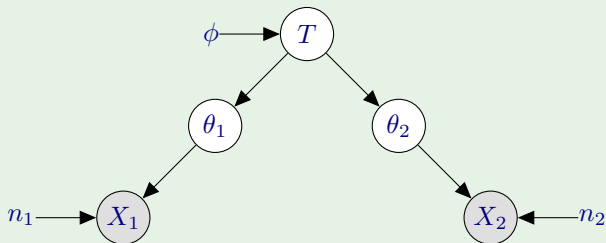
## Tossing a biased(?) coin

- $X$ : result of  $n$  coin tosses with some  $p(\text{heads})$
- random variables?
- fixed quantities?
- hidden variables?
- coin is possibly biased towards tails



## Some simple examples

### Two coins

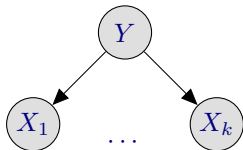




# Some simple examples

## Naive Bayes

- Predicting the value of categorical variable  $Y$  from a set of features  $X_1, \dots, X_k$



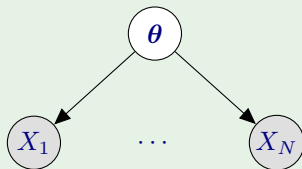
$$\begin{aligned} p(y \mid x_1, \dots, x_k) &\stackrel{\text{Bayes}}{=} \frac{p(x_1, \dots, x_k \mid y)p(y)}{p(x_1, \dots, x_k)} \\ &\propto p(x_1, \dots, x_k \mid y)p(y) = p(y) \prod_{i=1}^k p(x_i \mid y) \end{aligned}$$

# Plate notation

The idea is to avoid **repeated substructures**

## Example: independent data points

- Assume the elements in a sample  $X_1, \dots, X_N$  are independent if the parameter  $\theta$  is known



Unfolded notation

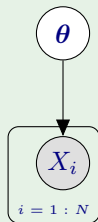


Plate notation

# Plate notation: linear regression revisited

## Example: linear regression (fully probabilistic)

- $Y_i \mid \{w, x_i\} = w^\top x_i + \epsilon_i$  with  $x_i = [1, x_i]^\top$
- $\epsilon_i \sim \mathcal{N}(0, 1/\gamma)$  with known precision parameter  $\gamma$
- $w \sim \mathcal{N}(\mu_0 = \mathbf{0}, \Sigma_0 = \mathbf{I}_{2 \times 2})$

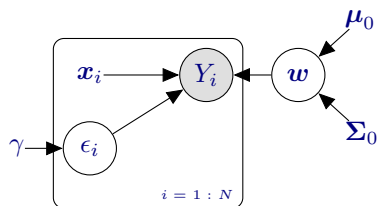
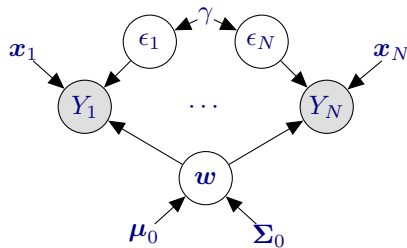


Plate notation



Unfolded notation

- Underlying model:  $y_i = w_0 + w_1 x_i + \epsilon_i$

# Generative and discriminative models

Predicting  $Y$  from  $X$

## Discriminative model

- Estimate  $p(y|x)$  directly from data

- Logistic regression, NNs, ...
- Lower asymptotic error but reached more slowly

## Generative model

- Learn  $p(x, y) = p(x|y)p(y)$  from data
- Compute  $p(y|x)$  using Bayes rule

- Naive Bayes, Bayesian networks in general, ...
- Can be used to generate synthetic data
- Higher asymptotic error but reached more quickly

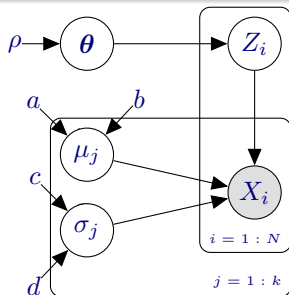
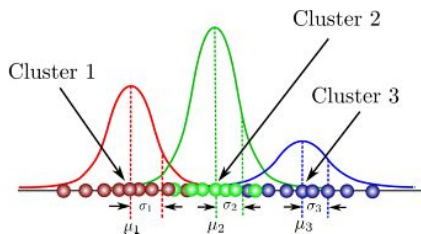
# Latent variable models

In general, **latent variable models** are regarded as probabilistic models where some of the variables cannot be observed.

## Example: mixture of Gaussians; popular model for clustering

Model formulation:

- $k$  Gaussians with frequencies  $\theta = (\theta_1, \dots, \theta_k)$  (sum to 1)
- $N$  observations generated by
  - $Z_i \sim \text{Multinom}(\theta)$ ,  $i = 1, \dots, N$  (indicates which Gaussian/cluster)
  - $X_i | z_i \sim \mathcal{N}(\mu_i, \sigma_i^2)$ ,  $i = 1, \dots, N$
  - Bayesian setting: priors on the parameters

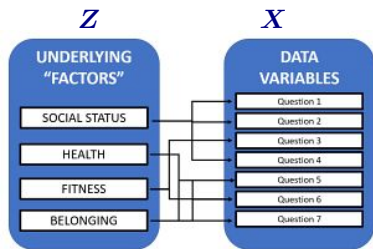


# Latent variable models

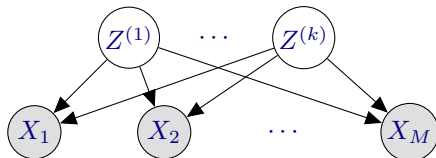
## Example: factor analysis (FA) model

FA summarizes a high-dimensional observation  $\mathbf{X}$  of  $M$  correlated variables by a smaller set of factors  $\mathbf{Z}$  assumed independent a priori.

Local model for single observation  $\mathbf{X}$



[https://golden.com/wiki/Factor\\_analysis-RWGG9](https://golden.com/wiki/Factor_analysis-RWGG9)



$$x_1 = w_{1,1} \cdot z^{(1)} + \dots + w_{1,k} \cdot z^{(k)} + \theta_1$$

$\vdots$

$$x_j = w_{j,1} \cdot z^{(1)} + \dots + w_{j,k} \cdot z^{(k)} + \theta_j$$

$\vdots$

$$x_M = w_{M,1} \cdot z^{(1)} + \dots + w_{M,k} \cdot z^{(k)} + \theta_M$$

# Latent variable models

## Example: factor analysis (FA) model

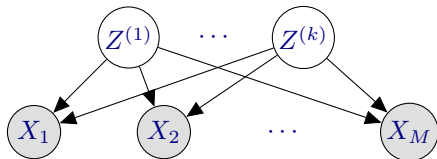
FA summarizes a high-dimensional observation  $\mathbf{X}$  of  $M$  correlated variables by a smaller set of factors  $\mathbf{Z}$  assumed independent a priori.

- Model formulation:

- $N$  observations  $\mathbf{X}_i$  generated by:

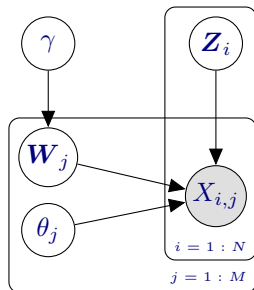
- ★  $\mathbf{Z}_i \sim \mathcal{N}(\boldsymbol{\mu}_0 = \mathbf{0}, \boldsymbol{\Sigma}_0 = \mathbf{I}), i = 1, \dots, N$
- ★  $X_{i,j} | \{\mathbf{z}_i, \mathbf{w}_j, \theta_j\} \sim \mathcal{N}(\mathbf{w}_j^\top \mathbf{z}_i, 1/\theta_j), i = 1, \dots, N, j = 1, \dots, M$
- ★  $\mathbf{W}_j \sim \mathcal{N}(\mathbf{0}, 1/\gamma)$

Local model for single observation  $\mathbf{X}_i$



$$x_{i,1} = w_{1,1} \cdot z_i^{(1)} + \dots + w_{1,k} \cdot z_i^{(k)} + \theta_1$$

etc.



# Exact inference

We've already seen Variable Elimination as an example:

$$p(x_5) = \sum_{x_2, \dots, x_4} p(x_4 | x_2, x_3) p(x_5 | x_4) h(x_2, x_3)$$

Considerations about exact inference:

- Product of functions raises complexity
  - Exponentially in the case of discrete variables
- Complexity also depends on the elimination order
- Representation of densities turns out to be relevant
  - Closed-form solutions to product and marginalization are preferable



# Approximate inference

- **sampling**: Monte Carlo techniques, e.g. importance sampling, MCMC
  - accurate with enough samples
  - sampling can be computationally demanding
- **deterministic**, e.g. variational approaches
  - uses analytical approximations to the posterior
  - some techniques scale well

# Monte Carlo inference algorithms

- A Bayesian network is a representation of a joint probability distribution over  $\mathbf{X} \Rightarrow$  it describes some **population** consisting of all the possible configurations of  $\mathbf{X}$
- If the entire population was available, the **inference problem** could be solved exactly, basically by **counting cases**
- **Problem:** Population size can be huge or even infinite.
- **Monte Carlo** methods operate by drawing an artificial **sample** from the population using some random mechanism
- The sample (**much smaller than the population**), is used to estimate the distribution of each variable of interest.

## Key issues in a Monte Carlo inference algorithm:

- ① The sampling mechanism
- ② The functions (estimators) which compute the probabilities from the sample

# Conclusions & Outlook

- PGMs provide a well founded way of handling uncertainty
- Bayesian POV: inference and learning are closely connected tasks
- If scalability is important, approximate inference is needed
- Interpretability is a key issue
- In addition to parameter learning for a given model/structure, the model itself can be learned from data



# Acknowledgements

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- Photo Peter Heine Nielsen: Wikimedia Commons (Stefan64)
- Andalusia map:  
[http://www.andalusia-web.com/map\\_nature\\_areas.htm](http://www.andalusia-web.com/map_nature_areas.htm)
- Bataille de Wagram (Napoleon): Wikimedia Commons (public domain)
- Tesla model 3: Getty Images